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RESEARCH ARTICLE

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Citizen science informs demand-driven breeding of opportunity crops

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Societal Impact Statement

Amid global challenges of food insecurity, poor nutrition, and climate change, neglected crops like amaranth are gaining renewed attention. We studied farmers' preferences for amaranth varieties across diverse geographical contexts to guide targeted breeding. Our results revealed significant variation in farmer preferences, emphasizing the need for context-specific breeding strategies. These findings can support the development of improved amaranth varieties that meet local needs, expand economic opportunities—especially for women—promote healthier diets, and boost biodiversity. This work also offers a model for participatory research on

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opportunity crops, informing inclusive agricultural policies and sustainable development strategies across Africa and beyond.

Summary

- Opportunity crops, also known as neglected and underutilized species (NUS), offer benefits to diversify food systems with nutritious and climate-resilient foods. A major limitation to incorporating these crops in farming systems is the lack of improved varieties, which impedes farmers from accessing quality planting materials of these crops. The study explored how citizen science methods can support demand-driven breeding and seed production of NUS using leafy amaranth – a nutritious and hardy vegetable - as a case study. The study identified farmer preferences and market segments, with particular attention to gender and social differentiation.
- We used the tricot approach to conduct participatory on-farm trials of 14 varieties with 2,063 farmers from Benin, Mali, and Tanzania. We then analyzed farmer traits and varietal preferences in aggregate and among segments of farmers, using cluster analysis.
- Farmers' overall preferences for amaranth varieties were driven principally by plant survival, yield, leaf size, taste, and marketability. Distinct farmer segments (older women generalists, young women specialists, older men generalists, and young men specialists) preferred different varieties depending on gender and business orientation.
- The identified farmer segments, along with their unique variety preferences, provide valuable information for breeders and seed enterprises, and support demand-driven amaranth breeding and seed system development. The methods used and lessons learned from our citizen science exercise can be applied to enhance breeding and seed supply of other opportunity crops that are underutilized in Africa and elsewhere.

KEYWORDS

African traditional vegetables, amaranth, data-driven agriculture, market segmentation, neglected and underutilized crops, socioeconomic heterogeneity, tricot approach

1 | INTRODUCTION

Supporting diversified farming systems that incorporate vegetables and fruits is an important response to challenges of malnutrition and climate vulnerability in Africa (Afshin et al., 2019; FAO et al., 2021; van Zonneveld et al., 2021, 2023; von Grebmer et al., 2014). Of particular interest are opportunity vegetables and fruits, of which many are currently neglected and underutilized (Mwadzingeni et al., 2021; van Zonneveld et al., 2023). These species are hailed for high levels of vitamins, minerals, antioxidants, and dietary fibers, their contributions to agrobiodiversity and climate resilience, and their suitability for African smallholder systems (Mwadzingeni et al., 2021; Schreinemachers et al., 2018; van Zonneveld et al., 2023; Yang & Keding, 2009).

Expanding the supply of these neglected and underutilized species (NUS) is a promising means to address nutrition challenges and boost smallholder incomes, as there is an increasing demand for these crops (Aworh, 2018; Dinssa et al., 2016; Okello et al., 2015; Tatsvarei & Rukasha, 2022). Bolstering NUS production and consumption, however, requires a range of interventions, including awareness-raising and demand creation among consumers, reforms to policies and subsidies to support NUS cultivation, agronomic advances to improve production, infrastructure development related to post-harvest handling, and expanded breeding to ensure NUS meet producer and consumer needs (Kalmpourtzidou et al., 2020; McMullin et al., 2021). These intervention points reflect the many factors underlying the underutilization of these crops, including persisting social stigma around their consumption and the deprioritization of NUS in policy, research, and development relative to current staple and cash crops (Kansiime et al., 2018; McMullin et al., 2021; Schreinemachers et al., 2018).

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Opportunity crops' potential to simultaneously address malnutrition, increase ecological sustainability of cropping systems and resilience to climate change, and preserve biodiversity has caused a renewed interest in these crops (Fredenberg et al., 2024). We focus here on breeding of NUS and farmer access to quality seed as critical components of expanded NUS supply to feed the growing demand in a sustainable way. However, limited availability and access often undermine the success of NUS production (Ayenan et al., 2021; McGuire & Sperling, 2016; McMullin et al., 2021). Since 2013, Africa's vegetable genebank maintained by the World Vegetable Center in Tanzania has distributed over half a million seed samples of promising genebank materials and breeding lines developed from genebank materials of a wide range of neglected and underutilized vegetables including amaranth, leafy nightshade, and jute mallow (N'Danikou et al., 2022). These efforts have increased adoption of these vegetables (Wanyama et al., 2023), but require further scaling because, so far, limited commercial offerings leave many farmers to cultivate varieties with lower yield potential or susceptible to climate change, pests, and diseases (Adebooye et al., 2005; Mwadzingeni et al., 2021; Schreinemachers et al., 2018).

Breeders and seed enterprises that might be interested in expanding NUS varietal offerings face capacity constraints and have little market intelligence to guide them (Afari-Sefa et al., 2012). Furthermore, varietal release processes and evaluation criteria (including Value for Cultivation and Use- VCU), through which new lines are tested and released, were designed principally for cereal crops. As such, they do not always measure characteristics of vegetables that are important to producers and consumers, such as color and shape, long seasonality, shelf life, texture, and taste (Afari-Sefa et al., 2012; Schreinemachers et al., 2021; Turner & Bishaw, 2016). Relevant traits for agroecological suitability, stress tolerance, and yield have therefore not been well identified or characterized for many NUS (Dinssa et al., 2016). This means that data to inform critical trade-offs in breeding between yield, abiotic and biotic stress tolerance, nutrition, commercially-relevant characteristics, and other traits are not readily available (Afari-Sefa et al., 2012).

In addition, little is known about gender-disaggregated preferences, demand, and willingness-to-pay for many traditional vegetables (summarized in Christinck et al., 2017; Weltzien et al., 2019; Wanyama et al., 2023). This knowledge gap could be a crucial oversight given women' heavy engagement in NUS production in rural areas, where they typically manage home gardens and prepare food for the household (Dinssa et al., 2016; Ojiewo et al., 2015). Commercial vegetable production, in contrast with subsistence production in home gardens, skews toward men (Wanyama et al., 2023; Weinberger & Pichop, 2009). However, women are primary marketers of NUS, even in cases where men are the primary producers (Dinssa et al., 2016; Fischer et al., 2020; Weinberger & Pichop, 2009). As such, men's and women's preferences as producers, marketers, and consumers of NUS must be better understood.

In this context, participatory breeding research is critical to ensure that promising accessions selected from NUS genebanks, and any improved varieties developed through them, respond to the real-world needs, constraints, and priorities of diverse farmers and consumers (Dinssa et al., 2022; van Etten et al., 2023; van Zonneveld et al., 2023). We used leafy amaranth as an example to develop a model to assess farmer preferences and market segments for NUS. Amaranth (*Amaranthus* spp.) is among the most commonly recognized traditional African vegetables, typically grown at a small scale and in home gardens (Ochieng et al., 2019). Amaranth is consumed primarily as a leafy vegetable in Africa, with demand for grain building (Schafleitner et al., 2022). Leaf nutrient content may vary with species and genotype. Most species constitute a good source of protein and calcium (particularly the grain), Vitamin C, iron, zinc, magnesium, and other minerals (Kachiguma et al., 2015; Kamga et al., 2013). Although WorldVeg seed kit distributions have helped disseminate improved varieties (Stoilova et al., 2019; Wanyama et al., 2023), access to quality amaranth seed remains a challenge, as there is not yet a wide diversity of improved amaranth varieties in many markets (Cernansky, 2015; Kansiime et al., 2018; Onim & Mwaniki, 2008).

2 | MATERIALS AND METHODS

2.1 | Trial design and variety evaluation

On-farm citizen science trials enabled participatory amaranth variety testing across a range of agroecological (humid coastal and drylands), socioeconomic (urban and peri-urban settings), and societal (cultural settings and gender) contexts. Trials were based on the triadic comparison of technologies (tricot) approach, in which individual farmers manage plots planted with a random subset of test varieties. Trial hosts evaluate each of their three varieties on multiple traits and at multiple stages in the growing season (van Etten et al., 2019, 2020). Data collection was conducted digitally using Open Data Kit (ODK), which was directly linked to the ClimMob platform (Quirós et al., 2024) for seamless aggregation and analysis. Survey data and protocols can be obtained via ClimMob (https://1000farms.climmob.net/) by selecting the amaranth protocol when designing a new trial. This design enables individual farmers to evaluate varietal performance in the context of their real-world needs and constraints, which aids in detecting variation in preference among types of producers, especially with regard to gender and social difference (Voss et al., 2021, 2023). Decentralized on-farm trials also mitigate some of the concerns that researcher-managed on-station and on-farm trials are not representative of farmers' actual growing conditions and are poor predictors of farmer preferences (De Roo et al., 2017; de Sousa et al., 2021, 2024; Laajaj et al., 2020). With this model, we sought to test a demand-driven participatory breeding approach that can be applied to other NUS as well as staple crops, and on larger scales.

Amaranth trials (Figure 1) were conducted with 794 farmers in the Atlantic, Oueme, and Alibori regions of Benin, and with 969 farmers in the Bougouni, Sikasso, and Koulikoro regions in Mali, from 2021 to 2022. Trials were also conducted with 300 farmers in Mtwara and Lindi regions in Tanzania in 2022. In all three countries,

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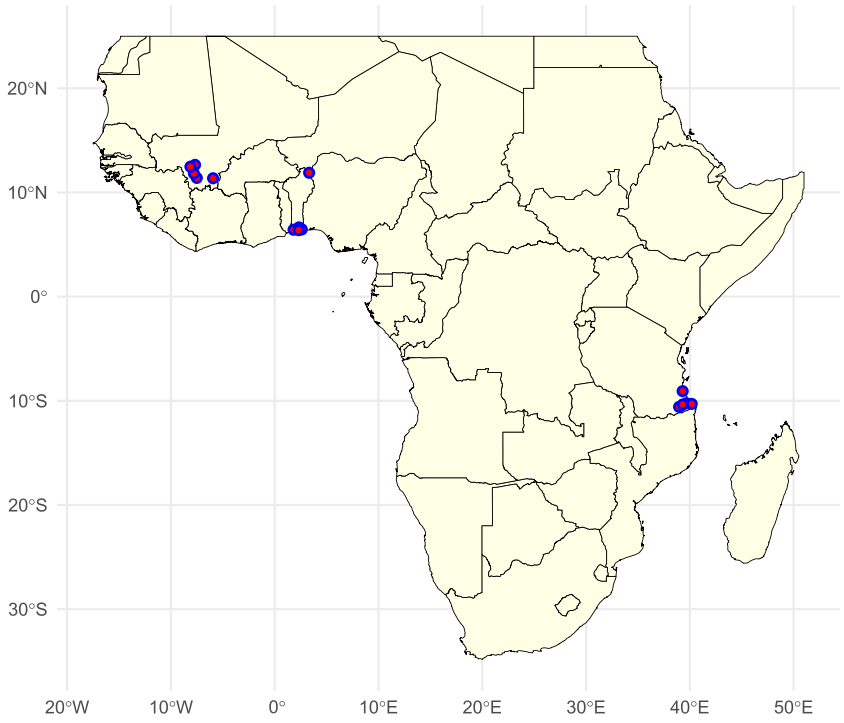


FIGURE 1 Centroids (blue-red dots) of amaranth on-farm trials in Benin, Mali, and Tanzania.

TABLE 1 Varieties distributed in trials, and number of farmers per country.

Genotype	Checks*	Seed source	Benin*	Mali*	Tanzania*
AC-NL		Genebank	233	324	100
A2004	Check in Mali	Genebank	148	225	
A2002		Breeding	148	323	
Akeri		Genebank	231	322	100
AM-NKGN		Breeding	234		
AVAM1938		Breeding			100
AVAM1939		Breeding			100
AVAM1941		Breeding			100
IP-5-Sel		Breeding	228	322	
Local	Check in Benin and Mali	Genebank	231	99	
Madiira 1	Check in Tanzania	Breeding	231	323	100
Madiira 2		Breeding	234	324	100
Nguruma		Breeding	232	324	100
Poli		Breeding	232	321	100

*Blank cells mean that the variety was not part of the portfolio of varieties tested in that particular country.

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trials used a balanced incomplete block design under which host farmers received three of fourteen amaranth genotypes and promising accessions drawn from the WorldVeg genebank (Table 1). Each farmer received 2 g of seed per variety, with variety names coded A, B, and C. Plot sizes (5 × 2 m in Mali and Tanzania, 6 × 1.2 m in Benin) and plant spacing (60 × 40 cm in Mali, 20 × 20 cm in Benin, 15 × 15 cm in Tanzania) were recommended but not strictly enforced. Farmers were permitted to practice their preferred management so long as it was consistent across their three plots.

Farmers evaluated their three varieties throughout the season across four harvest periods. Evaluations involved individual farmer's ranking varieties' overall performance and specific agronomic and end-use traits: germination, vigor, plant survival, pest tolerance, disease resistance, drought and flood tolerance, plant height, branching, yield, leaf size, marketability, and taste (as consumers of their own products). The socio-economic data on the host farm and farmers were collected through a structured baseline survey targeting the household member responsible for

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managing the amaranth trial. The survey captured key indicators, including demographic characteristics (gender, age, education level, literacy, household type, and size), land use and production factors (land ownership, access, availability for expansion, irrigation access, and type of irrigation system), and crop management practices (experience with amaranth, production frequency, cultivated area, main vegetable crops, seed sources, and farming techniques such as row planting, use of fertilizers, and seed-saving practices). Market-related data included the share of amaranth production sold, its contribution to household income, selling methods, and distance to markets. Institutional and social aspects were also covered, including access to government and private extension services, membership in farmer or savings groups, and household assets. Data collection was conducted digitally using ODK and linked to the ClimMob platform, allowing real-time entry and minimizing errors associated with manual recording.

2.2 | Data analysis

We analyzed the tricot ranking data using the Plackett-Luce model and model-based recursive partitioning (Luce, 1959; Plackett, 1975; Zeileis et al., 2008), recommended for analysis of on-farm tricot data (de Sousa et al., 2021). This model produces scaleless, quantitative estimates of individual varietal performance for different traits, reflecting the probability of each variety outperforming all other varieties in the tested set as log-*worth* estimates. Due to the large number of traits assessed (10), some over multiple growth stages in the season (up to six data collection moments), we used Kendall Tau correlation to identify the traits most closely associated with farmers' overall preference for the tested varieties. Traits were selected using a backward selection approach, starting with the full set of traits and iteratively removing the traits with the least correlation to overall preference until no uncorrelated traits remain ($p > 0.05$). These traits were used to perform the likelihood-ratio test (described below) and principal component analysis with Plackett-Luce coefficients obtained for each trait. For the modelling and data processing, we used the R packages *PlackettLuce*, *ClimMobTools*, and *gosset* (de Sousa et al., 2023; de Sousa & van Etten, 2024; Turner et al., 2020). We used principal components analysis to visualize the relationship between tested varieties and the traits assessed by farmers by plotting the Plackett-Luce coefficients computed for each trait.

To identify potential producer segments, we applied a cluster analysis using the farmers' socioeconomic data. This was done following existing producer segmentation studies and independently from the variety performance rankings (Hammond et al., 2020; Kilwinger et al., 2021). We used covariates relevant to producer preferences identified in past studies and reflective of grower and end-user requirements in seed product market segments (Donovan et al., 2022). The variables used were gender, age, years of experience growing amaranth, distance to markets, gender of individual controlling production, gender of individual controlling selling, and household income share from amaranth production (Tables 2 and 3). Segmentation was performed in R using the package *cluster* (Maechler et al., 2023). Categorical variables were converted to factors and dissimilarities computed using a pairwise dissimilarity approach. Numeric variables were standardized, and Euclidean distances were calculated. The resulting distance matrices underwent hierarchical clustering, and optimal clusters were determined to retain distinguishable clusters.

TABLE 2 Characteristics of trial participants and their households/farms.

	Overall	Benin	Mali	Tanzania	Men	Women
Trial participants	2063	794	969	300	904	1,159
Men	43.8%	66.9%	16.9%	69.7%		
Women	56.2%	33.1%	83.1%	30.3%		
Average household size	11	16.7	18.4	3.9		
Household structure						
Nuclear family	51.6%	73.9%	22.7%	67.7%	65.4%	39.9%
Polygamous family	32.1%	18.5%	58.0%	3.3%	20.5%	41.9%
Single-parent household	16.4%	7.6%	19.3%	29.0%	14.2%	18.2%
Average share of amaranth crop sold (among households that sell)	58.1%	81.0%	40.2%	54.8%	63.2%	54.0%
Average share of HH income made through amaranth sales (among households that sell)	24.7%	22.1%	24.0%	34.2%	21.4%	27.3%
Primary employment						
Farming	93.6%	94.8%	93.5%	91.3%	91.3%	95.6%
Salaried employment	1.7%	0.8%	2.5%	1.7%	2.7%	0.8%
Self-employment off-farm	3.8%	2.7%	3.4%	7.0%	4.9%	2.8%
Other	0.9%	1.7%	0.6%	0.0%	1.1%	0.8%
Farmers using irrigation on farm	84.9%	83.6%	83.1%	92.0%	87.4%	82.8%
Engagement with extension services	67.9%	85.8%	45.1%	80.0%	79.4%	58.2%

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TABLE 3 *Household-level control over amaranth production, sales, and seed exchange, as reported by the trial participants.*

	Overall	Benin	Mali	Tanzania
Control over amaranth production				
Man	40.0%	58.9%	13.2%	60.0%
Woman	48.2%	29.3%	75.1%	27.7%
Both	8.4%	4.0%	11.4%	11.0%
Other	3.4%	7.8%	0.3%	1.3%
Control over amaranth sales				
Man	34.4%	50.8%	7.5%	59.0%
Woman	52.0%	32.4%	81.4%	28.7%
Both	7.8%	4.7%	9.5%	10.7%
Other	5.8%	12.1%	1.7%	1.7%
Control over amaranth seed exchange				
Man	39.8%	56.2%	14.4%	61.3%
Woman	46.3%	28.5%	71.1%	28.0%
Both	10.1%	5.5%	14.4%	9.3%
Other	3.9%	9.8%	0.2%	1.3%

Subsequent refinement and validation of clusters were conducted, leading to the identification of four distinct clusters (segments). After the definition of segments in R, we used ChatGPT (OpenAI, San Francisco, USA) to assist in generating a narrative description of the main characteristics of each segment using the full set of covariates. The idea was to describe these segments from a persona-based approach. These descriptions were reviewed by the authors and edited to ensure segment names and descriptions were consistent with the data and with each other.

We then used a likelihood-ratio test (*likelihood_ratio* function in the R package *gosset*) to assess whether varietal rankings for key traits retained by the Kendall correlation differed significantly between different segments. Briefly, the Plackett-Luce model is fitted using maximum likelihood, which allows the log-likelihood for a single model fitted to the full dataset to be compared to the sum of log-likelihoods for separate models fitted to pre-defined splits of the data, accounting for the increase in degrees of freedom in such a segmented model. Principal components analysis was also performed to visualize the relationship between traits and varieties in each segment.

Finally, we performed a regret analysis (*regret* function in the R package *gosset*) using coefficients from the Plackett-Luce rankings of marketability. Regret is a risk assessment analysis to support farmers' diversification analysis. We calculated minimum regret by taking the summed squares of the distances of each variety to the best variety in each market segment and taking the square root of the resulting sum. It can be interpreted as the total distance to the 'best variety' in each group using the log-worth estimates. This measure is therefore more sensitive to differentiated preferences than the overall worth, which could be biased if one group has a strong preference for a particular variety that is less preferred by other segments.

3 | RESULTS

3.1 | Sample characteristics

Across countries, 56.2% of the trial participants were women, although men were disproportionately represented in Tanzania (69.7%) and Benin (66.9%) and under-represented in Mali (16.9%) (Table 2). This may reflect the higher degree of commercialization of amaranth in Benin and Tanzania. Roughly half of participating households were nuclear families, although the majority of women trial participants came from either polygamous or single-parent households.

Farming was the dominant occupation across countries. Among the 84% of households that reported selling amaranth, on average, 58% of amaranth produced was sold (81% in Benin). These sales contributed 25% of household income on average among households selling amaranth.

Table 3 suggests that women are slightly more engaged than men in amaranth activities overall, and especially sales. However, there is substantial variation in men's and women's engagement between countries. While amaranth production, marketing, and seed exchange were reportedly primarily undertaken by men in Benin and Tanzania, women in Mali held disproportionate responsibility for amaranth. These differences are, at least in part, likely a result of the gender balance of the trials in the three countries, as both women and men were more likely to report themselves as having control over amaranth than to report that their partner has control.

3.2 | Overall trait and variety preferences

The Kendall-Tau correlations showed that plant survival (both during the reproductive phase and at the third harvest), yield (primarily at later harvests), taste and leaf size (at final harvest), and marketability (at all harvest periods) were the traits that drove farmers' overall preferences (Figure 2). This allowed us to focus on these traits as those most relevant to farmers' overall choices.

In aggregate, overall variety preferences skewed toward Akeri, Poli, and AM-NKGN, linked to their marketability, leaf size, yield, and taste at the fourth harvest (Figure 3). These varieties were often reported to be preferred for use in subsequent seasons. A2004 emerged as another popular variety, driven by its marketability in the first three harvests, yield, and plant survival in the third harvest, but less by overall preference.

3.3 | Variation in farmer preferences

To consider variation in farmer preferences, we undertook a segmentation process that produced four farmer segments (Table 4). These segments are distinguished by gendered control over amaranth, income generated, and experience in amaranth farming. The "Older Women Generalists" segment represents women who have significant control over both the sale and production of amaranth. They earn only

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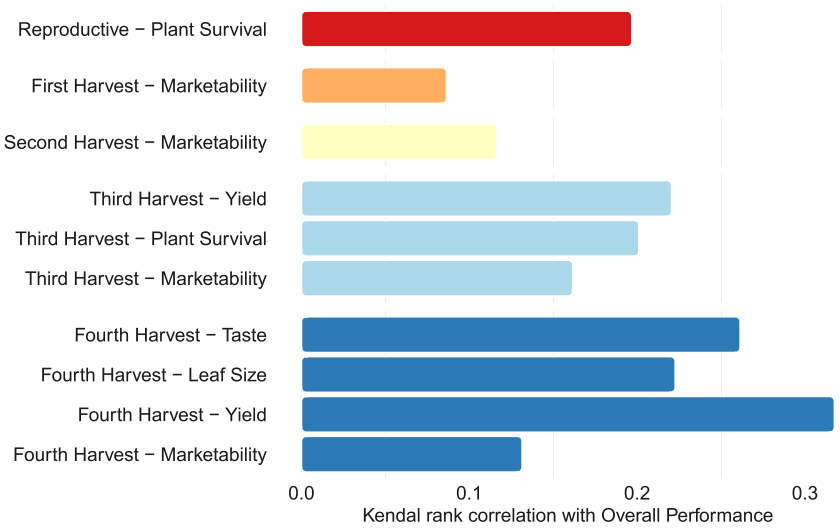


FIGURE 2 Among the many traits which farmers evaluated throughout the season, marketability, yield (especially at later harvests), taste, plant survival, and leaf size were most strongly correlated with farmers' overall variety preferences. Traits are sorted out based on crop stages.

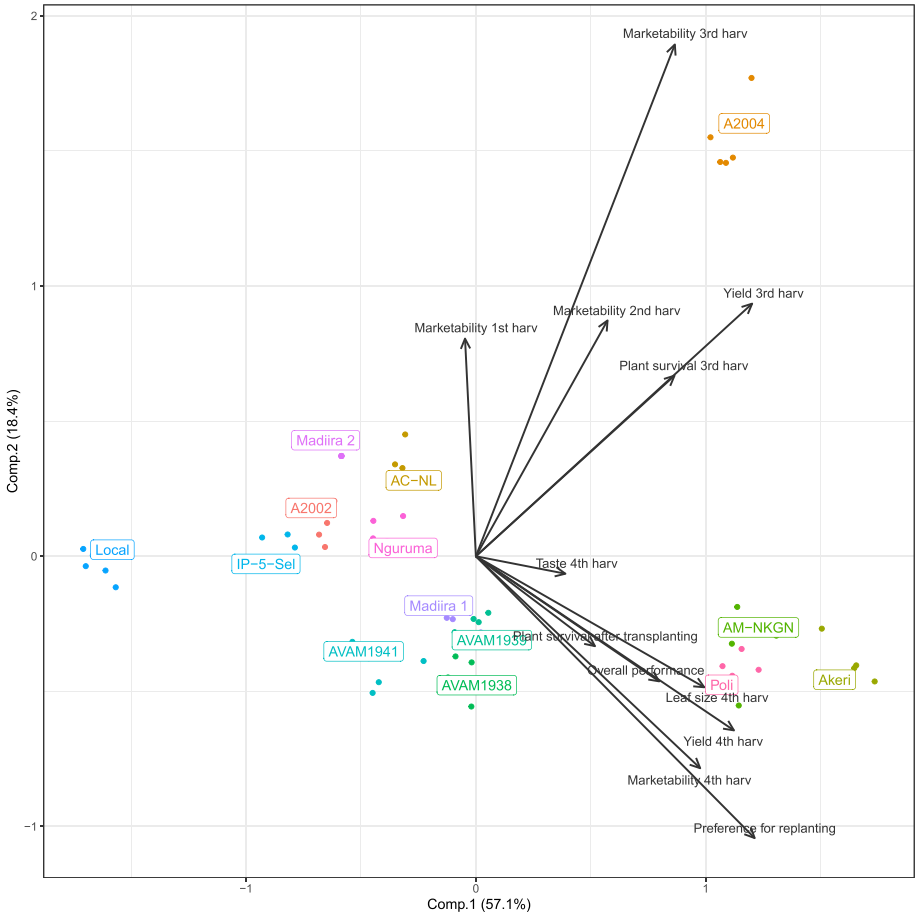


FIGURE 3 Principal components of Plackett-Luce model estimates (log-worth) on amaranth trait performance. Dots represent the performance (log-worth) of each amaranth variety ranked by farmers. Arrows represent the paths (correlation) of varieties and the main traits retained after backward selection.

a moderate share of income from amaranth and have relatively less experience in amaranth farming. "Young Women Specialists" includes younger women who were highly involved in both the production and sale of amaranth. They boasted the highest income share from amaranth and substantial experience cultivating it, indicating a greater degree of specialization in amaranth. "Older Men Generalists" are predominantly men with considerable experience in amaranth farming. They have control over both the production and sale of amaranth, but a lower income share from the crop. The "Young Men Specialists" segment represents younger men with the highest average experience cultivating amaranth and a significant income share from amaranth. They controlled both the production and sales.

To validate these groups, we conducted a log-likelihood ratio test. Table 5 indicates whether each segment generated statistically different rankings on the key traits retained by the Kendall partial correlation. Other than traits ranked at the fourth harvest (when a smaller

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TABLE 4 *Demographic and socioeconomic characteristics of farmers segments in amaranth production in Benin, Mali, and Tanzania.*

	Segments			
	Older women generalists	*Young women specialists*	*Older men generalists*	*Young men specialists*
Average age (years)	43	35	45	34
Dominant gender	Woman	Woman	Man	Man
Who controls Sale	Woman	Woman	Man	Man
Who controls production	Woman	Woman	Man	Man
Avg. income crop share (%)	21.59	35.42	14.17	29.01
Avg. experience with crop (years)	1.18	6.2	5.52	8.95
Characteristics	Women with significant control over both sale and production, moderate income share from amaranth, relatively less experience in amaranth farming.	Younger women highly involved in both production and sale, highest income share from amaranth, and substantial experience growing amaranth, indicating specialization.	Predominantly men with considerable experience in amaranth farming, control over both production and sale, but the lowest average income share from the crop.	Younger men with high experience, significant income share from amaranth, control over both production and sale, indicating specialization.

TABLE 5 Log-likelihood ratio test estimates for the main traits assessed by farmers within segments.

Trait	Deviance	Pr(>Chisq)	
Plant survival after transplanting	65.239	0.005	*
Marketability 1st harvest	57.684	0.027	*
Marketability 2nd harvest	59.875	0.017	*
Plant survival 3rd harvest	70.380	0.002	*
Yield 3rd harvest	59.204	0.020	*
Marketability 3rd harvest	54.843	0.048	*
Yield 4th harvest	45.576	0.217	
Leaf size 4th harvest	35.902	0.612	
Taste 4th harvest	49.284	0.125	
Marketability 4th harvest	40.070	0.422	
Overall performance	48.193	0.148	

*Indicates a significantly different ranking based on the particular trait.

number of observations were recorded), all key traits' rankings were distinguished across the four segments.

Figure 4, in contrast with Figure 3's aggregated model, shows that trait and variety preferences differed across the four farmer segments. Older Women Generalists expressed a strong preference for Akeri, Poli, and AVAM1941, and a dislike of the other varieties (those in the western quadrants of Figure 4A). Young Women Specialists similarly identified five clearly preferred varieties (A2004, Akeri, AVAM1939, AM-NKGN, and Poli) and disliked the remainder. Older Men Generalists' variety preference model shows a much more even spread of arrows and variety clusters, indicating little agreement about optimal varieties. Young Men Specialists' model indicates strong preferences for A2002 and A2004 varieties driven by their marketability, while a large number of other varieties (Madiira 1, Nguruma, AC-NL, and Akeri) were preferred for other traits such as yield, leaf size, and plant survival.

Finally, to provide a synthetic analysis to inform decision making, we present both worth and minimum regret values for each of the varieties (Table 6). This measure gives an indication of the 'loss' that would be perceived by the different segments in using one variety rather than the top-ranked variety for each group. Independent from the measure taken, worth or regret, A2004 would be acceptable to all groups. However, if a second variety is to be recommended, AC-NL could be chosen based on its overall high worth, but Poli would minimize regret. Choosing AC-NL would mainly benefit Older Men Generalists, whereas Poli is preferred by the three other groups, and therefore, a more balanced choice.

4 | DISCUSSION

4.1 | Implications for demand-driven amaranth breeding

Our results provide useful information for amaranth breeders and seed producers to select varieties and traits for specific market segments. First, our results point to a distinct segmentation of amaranth farmers based on gendered control of amaranth production

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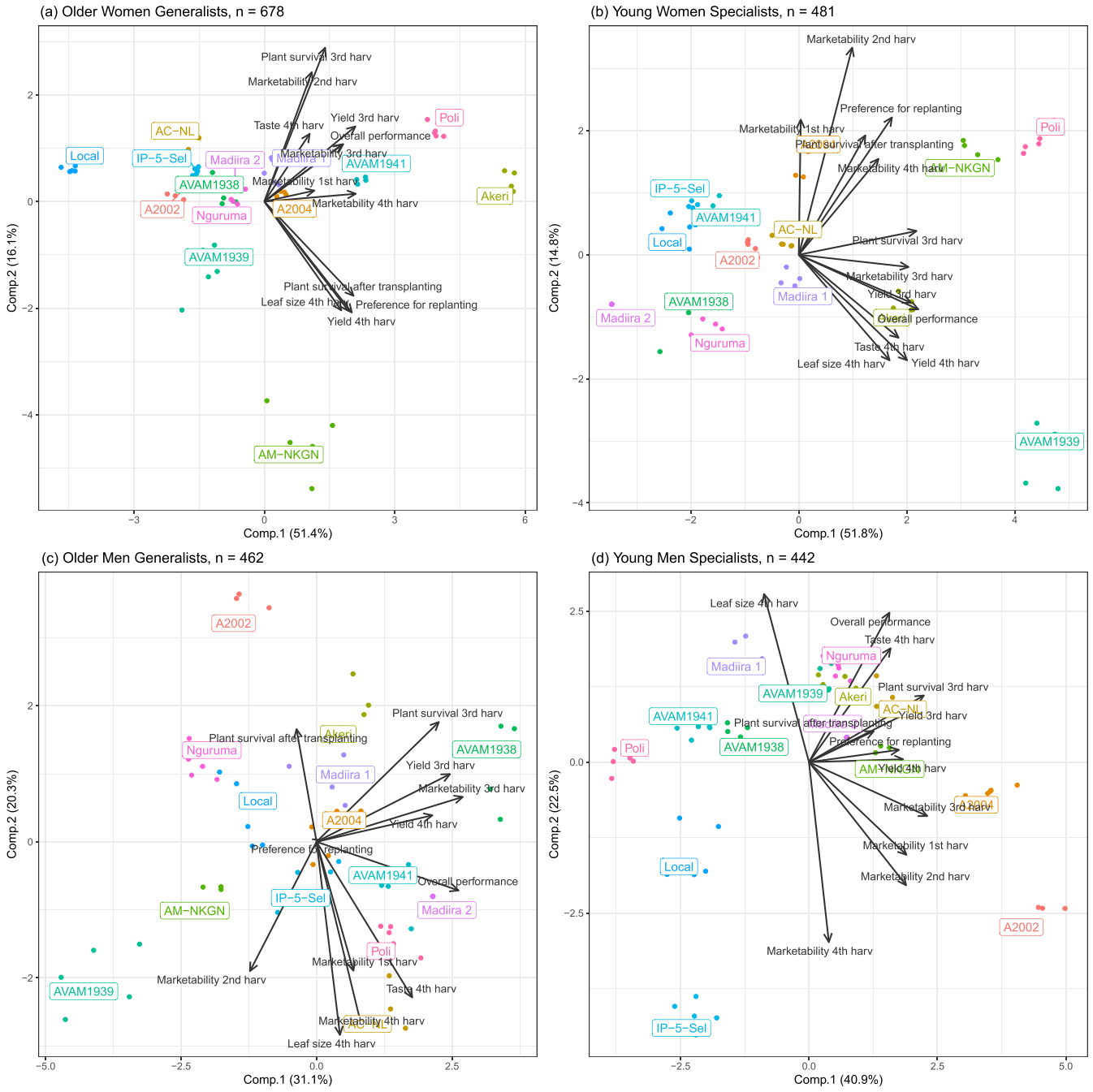


FIGURE 4 Principal components of Plackett-Luce model estimates (log-worth) on farmers' segments. Each panel represents the performance of amaranth varieties by trait within the farmer segments identified through cluster analysis. The arrows indicate the direction and strength of the correlation between traits and variety performance within each segment, while the dots represent individual variety rankings based on the Plackett-Luce model. Performance of amaranth varieties for the older women generalist market segment (panel A), performance of amaranth varieties for the young women specialists segment (panel B), performance of amaranth varieties for the older men generalists segment (panel C), performance of amaranth varieties for the young men specialists segment (panel D).

and sales, income generation, and experience producing amaranth. Variety preferences differed across these segments, underscoring the heterogeneity of NUS producers. Second, our results also provide specific insights into trait preferences to guide breeding programs. Farmers' overall preferences for amaranth are related primarily to yield, taste, plant survival from early to late stages of the season, leaf size, and marketability. These bear a resemblance to priority traits documented in other studies in Tanzania (Adeniji & Aloyce, 2013; Dinssa et al., 2022). In a recent study in Tanzania, for instance, both women and men farmers ranked (1) fast growth habit (early biomass accumulation) plus quick recovery from repeat harvests, (2) marketability, and (3) ability to be harvested several times from the same plants as the three most important traits (Dinssa et al., 2022). With these insights and specific variety preferences,

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TABLE 6 Average worth, minimum regret values, and standard errors (SE) for the trait 'marketability' of evaluated amaranth varieties. Values in bold highlight the three varieties with smaller minimum regret.

	Worth		Minimum regret	
Variety	Value	SE	Value	SE
A2004	0.2317	0.0840	**0.0011**	0.0008
AC-NL	0.0732	0.0139	0.0725	0.0364
Poli	0.0694	0.0127	**0.0631**	0.0292
Madiira 2	0.0687	0.0193	0.0788	0.0401
A2002	0.0684	0.0251	0.0786	0.0392
Akeri	0.0649	0.0139	**0.0710**	0.0384
AVAM1941	0.0638	0.0156	0.0715	0.0376
AM-NKGN	0.0616	0.0098	0.0748	0.0358
Local	0.0553	0.0133	0.0812	0.0388
IP-5-Sel	0.0551	0.0101	0.0789	0.0385
AVAM1939	0.0485	0.0020	0.0786	0.0369
Nguruma	0.0484	0.0096	0.0838	0.0410
AVAM1938	0.0464	0.0102	0.0833	0.0403
Madiira 1	0.0446	0.0059	0.0829	0.0394

breeders can prioritize new crosses to meet current and future demand.

These results show the potential of citizen science to enhance decision-making in NUS breeding more generally. First, it helps to identify the target product profiles for different market segments, following the trait preferences of growers in these different segments. It also helps to identify which varieties and genebank accessions contain promising traits for breeding and selection. Third, it allows for which varieties are most useful for which growers and in which geographic areas. Furthermore, it enables testing of which markets are suitable for specific NUS crops and varieties. Finally, it can accelerate variety release by integrating Tricot trials results with DUS (Distinctiveness, Uniformity, and Stability) tests.

Taste emerged as a key factor driving farmers' preferences, but it is not consistently included in breeding programs as a priority trait. Organoleptic properties' historic exclusion from varietal testing and release processes (Afari-Sefa et al., 2012; Schreinemachers et al., 2021; Turner & Bishaw, 2016) and limited consumer preference research in breeding programs (Thiele et al., 2020) are likely one reason for this. Infamously, tomato breeding in the Netherlands had to drastically switch course after the market collapsed due to consumer dissatisfaction with tasteless tomatoes (Schouten et al., 2019). Breeding programs of neglected and underutilized vegetables should account for consumer taste from the start to avoid this risk. Increased attention to taste may have particular relevance for gender-responsive breeding and seed systems development, given evidence that women disproportionately prioritize these traits across crops (Weltzien et al., 2019). Taste is associated with phytonutrient concentrations and corresponding health benefits (Hounsome et al., 2008; Jarvis et al., 2024). It requires further participatory research and screening on phytonutrient concentrations and their contributions to flavor and health.

4.2 | Gender implications

The results underscore the relevance of gender considerations in amaranth breeding, with implications for wider NUS breeding. The dynamics of intrahousehold control of amaranth production, sales, and seed exchange (Table 3) indicate that both women and men are (or at least perceive themselves to be) involved in amaranth-related activities. This aligns with a previous study in Tanzania that found production activities for leafy vegetables to be shared, although they identified seed selection for vegetables to be largely men's responsibility (Fischer et al., 2017). Understanding and appealing to both men's and women's needs and constraints in breeding is critical, especially when variety preferences differ, as found here. Attention to the gender dynamics of seed selection and acquisition is also warranted to ensure men's and women's ability to equitably *access* and *benefit from* improved varieties.

Our results contrast with those from a conventional, gender-disaggregated participatory varietal selection study of amaranth in Tanzania, where female and male farmers' variety preferences were found to be similar (Dinssa et al., 2022). This discrepancy likely results in part from our use of a citizen science-based approach focused on understanding producer and consumer preferences grounded in men's and women's realities (Voss et al., 2023). Our analysis using market segments rather than gender-based disaggregation also illustrates how preferences may vary among women and among men according to their production orientation, experience, livelihood portfolio, and other factors. Such intersectional analyses of market segments are likely to yield deeper insights into preferences than gender-disaggregated varietal selection (e.g., Dinssa et al., 2022).

To illustrate this point, it is worth noting that Older Men Generalists' variety preferences were inconsistent and not well aligned with the other segments. This could simply reflect disagreement within this segment, but also suggests that these farmers are less certain than other segments about trait priorities. This could result from a lack of expertise in the Older Men Generalists group, perhaps linked to women's greater involvement in amaranth activities in many rural contexts; and men may have overstated their own role in amaranth cultivation in this study. This possibility is concerning given that older men's voices are often elevated in decision-making around topics like breeding. Ensuring that the preferences of younger amaranth specialists and older women are adequately captured may be key to appropriately meeting current and future seed demand.

4.3 | Supporting expanded production and consumption of NUS

Our results enable, perhaps for the first time, seed producers to focus NUS development on defined market segments, and specifically

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women and youth—two groups that are often trained and supported in NUS production. With more breeding and quality NUS seed supply, producers may be able to access more locally-adapted, climate-resilient, pest- and disease-tolerant, and nutritious varieties (Kansiime & Mastenbroek, 2016; Schreinemachers et al., 2021). While our findings can help expand NUS breeding and improve seed access, efforts to improve the appeal of NUS varieties for producers must ultimately be paired with attention to consumer demand and policy changes (McMullin et al., 2021). Local knowledge around utilization of these crops, breeding for consumer preferences, and improved post-harvest handling are all critical (Keatinge et al., 2015; Schreinemachers et al., 2018). There is also a need for value chain development that helps producers—and seed enterprises—to profit from NUS seed production and NUS sales (Onim & Mwaniki, 2008). This includes attention to postharvest processes and infrastructure to enable proper handling and storage of perishable vegetable products (Keatinge et al., 2015; Schreinemachers et al., 2021).

This study provides a model for using NUS for which little breeding work has thus far been done, and where farmers can evaluate traditional varieties from different regions and accessions maintained in genebanks. This is particularly timely because of the increased interest in breeding neglected vegetables (Fredenberg et al., 2024). In this way, citizen science can enhance the use of the intraspecific diversity in many NUS in national and international genebank collections, which will be relevant in the light of climate change, as it provides more options for producers to adapt their farming systems and crop combinations to climate change.

AUTHOR CONTRIBUTIONS

SN, KdS, and MvZ conceptualized the research; MvZ, SN, and AR acquired funds; SN, FD, AS, LA, ECL, ACH, DYD, AD, AS, COO, HC, EAD, AK, DM, and ZM coordinated data collection; KdS, MAL, and JvE performed analysis and developed figures; RCV, KdS, SN, and MvZ developed the first draft. All authors reviewed and contributed to the first and later drafts.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The survey template, data, and code that support the findings of this study are openly available on "Zenodo", at https://doi.org/10.5281/zenodo.15204183 (N'Danikou et al., 2025).

ETHICS STATEMENT

The study followed the Code of Ethics of the Society of Ethnobiology. The study was reviewed and approved by the Institutional Biosafety and Research Ethics Commission (IBREC) of the World Vegetable Center (Registration nr. 2022–08). Prior informed consent was obtained from all the participants before the start of the survey.

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REFERENCES

Adebooye, O. C., Ajayi, S. A., Baidu-Forson, J. J., & Opabode, J. T. (2005). Seed constraint to cultivation and productivity of African indigenous leaf vegetables. *African Journal of Biotechnology*, *4*(13), 1480–1484. https://www.ajol.info/index.php/ajb/article/view/71614/60584

Adeniji, O. T., & Aloyce, A. (2013). Farmers' participatory identification of horticultural traits: Developing breeding objectives for vegetable

25722611, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1002/ppp3.70035 by Norwegian Institute Of Public Healt Invoice Receipt DFO, Wiley Online Library on [15/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Amaranth in Tanzania. *Journal of Crop Improvement*, *27*(3), 309–318. https://doi.org/10.1080/15427528.2013.768318

Afari-Sefa, V., Tenkouano, A., Ojiewo, C. O., Keatinge, J. D. H., & Hughes, J. d'A. 2012). Vegetable breeding in Africa: Constraints, complexity and contributions toward achieving food and nutritional security. *Food Security*, *4*(1), 115–127. https://doi.org/10.1007/s12571-011-0158-8

Afshin, A., Sur, P. J., Fay, K. A., Cornaby, L., Ferrara, G., Salama, J. S., Mullany, E. C., Abate, K. H., Abbafati, C., Abebe, Z., Afarideh, M., Aggarwal, A., Agrawal, S., Akinyemiju, T., Alahdab, F., Bacha, U., Bachman, V. F., Badali, H., Badawi, A., … Murray, C. J. L. (2019). Health effects of dietary risks in 195 countries, 1990–2017: A systematic analysis for the global burden of disease study 2017. *The Lancet*, *393*(10184), 1958–1972. https://doi.org/10.1016/S0140-6736(19)30041-8

Aworh, O. C. (2018). From lesser-known to super vegetables: The growing profile of African traditional leafy vegetables in promoting food security and wellness. *Journal of the Science of Food and Agriculture*, *98*(10), 3609–3613. https://doi.org/10.1002/jsfa.8902

Ayenan, M. A. T., Aglinglo, L. A., Zohoungbogbo, H. P. F., N'Danikou, S., Honfoga, J., Dinssa, F. F., Hanson, P., & Afari-Sefa, V. (2021). Seed Systems of Traditional African Vegetables in eastern Africa: A systematic review. *Frontiers in Sustainable Food Systems*, *5*(September), 1–12. https://doi.org/10.3389/fsufs.2021.689909

Cernansky, R. (2015). Super vegetables. *Nature*, *146–148*, 146–148. https://doi.org/10.1038/522146a

Christinck, A., Weltzien, E., Rattunde, F., & Ashby, J. A. (2017). *Gender differentiation of farmer preferences for varietal traits in crop improvement: Evidence and issues*. In *Working Papers* (No. 2). CGIAR Gender and Agriculture Research Network; CGIAR System Management Office and International Center for Tropical Agriculture (CIAT). Cali, . 38 p. https://cgspace.cgiar.org/server/api/core/bitstreams/2ca274bb-cc07-4dfc-86c1-6930144b3056/content

De Roo, N., Andersson, J. A., & Krupnik, T. J. (2017). On-farm trials for development impact? The organisation of research and the scaling of agricultural technologies. *Experimental Agriculture*, *55*(2), 163–184. https://doi.org/10.1017/S0014479717000382

de Sousa, K., Brown, D., Steinke, J., & van Etten, J. (2023). Gosset: An R package for analysis and synthesis of ranking data in agricultural experimentation. *SoftwareX*, *22*, 101402. https://doi.org/10.1016/j.softx.2023.101402

de Sousa, K., & van Etten, J. (2024). ClimMobTools: API Client for the "ClimMob" Platform. R package *v1.5*.https://doi.org/10.32614/CRAN.package.ClimMobTools

de Sousa, K., van Etten, J., Manners, R., Abidin, E., Abdulmalik, R. O., Abolore, B., Acheremu, K., Angudubo, S., Aguilar, A., Arnaud, E., Babu, A., Barrios, M., Benavente, G., Boukar, O., Cairns, J. E., Carey, E., Daudi, H., Dawud, M., Edughaen, G., … Zaman Allah, M. (2024). The tricot approach: An agile framework for decentralized on-farm testing supported by citizen science. A retrospective. *Agronomy for Sustainable Development*, *44*(8), 1–19. https://doi.org/10.1007/s13593-023-00937-1

de Sousa, K., van Etten, J., Poland, J., Fadda, C., Jannink, J. L., Kidane, Y. G., Lakew, B. F., Mengistu, D. K., Pè, M. E., Solberg, S. Ø., & Dell'Acqua, M. (2021). Data-driven decentralized breeding increases prediction accuracy in a challenging crop production environment. *Communications Biology*, *4*(1), 944. https://doi.org/10.1038/s42003-021-02463-w

Dinssa, F. F., Hanson, P., Dubois, T., Tenkouano, A., Stoilova, T., Hughes, J. A., & Keatinge, J. D. H. (2016). AVRDC – The world vegetable center's women-oriented improvement and development strategy for traditional African vegetables in sub-Saharan Africa. *European Journal of Horticultural Science*, *81*(2), 91–105. https://doi.org/10.17660/eJHS.2016/81.2.3

Dinssa, F. F., Minja, R., Kariuki, T., Mbwambo, O., Schafleitner, R., & Hanson, P. (2022). Gender-disaggregated farmers participatory variety selection in Amaranth multilocation trials in Kenya and Tanzania. *HortTechnology*, *32*(3), 288–303. https://doi.org/10.21273/HORTTECH04963-21

Donovan, J., Coaldrake, P., Rutsaert, P., Bänziger, M., Gitonga, A., Naziri, D., Demont, M., Newby, J., & Ndegwa, M. (2022). Market intelligence for informing crop-breeding decisions by CGIAR and NARES. *Market Intelligence Brief Series*, *1*, 8. https://hdl.handle.net/10883/22248

van Etten, J., Abidin, E., Arnaud, E., Brown, D., Laporte, M., López-noriega, I., Manners, R., Ortiz-crespo, B., Quirós, C., De, K., Teeken, B., Tufan, H. A., Ulzen, J., & Valle-soto, J. F. (2020). *The tricot citizen science approach applied to on-farm variety evaluation: methodological progress and perspectives* (No. 2; RTB Working Paper). https://doi.org/10.4160/23096586RTBWP20212

van Etten, J., de Sousa, K., Aguilar, A. A., Barrios, M., Coto, A., Dell'Acqua, M., Fadda, C., Gebrehawaryat, Y., van de Gevel, J., Gupta, A., Kiros, A. Y., Madriz, B., Mathur, P., Mengistu, D. K., Mercado, L., Mohammed, J. N., Paliwal, A., Pè, M. E., Quirós, C. F., … Steinke, J. (2019). Crop variety management for climate adaptation supported by citizen science. *Proceedings of the National Academy of Sciences of the United States of America*, *116*(10), 4194–4199. https://doi.org/10.1073/pnas.1813720116

van Etten, J., de Sousa, K., Cairns, J. E., Dell'Acqua, M., Fadda, C., Guereña, D., van Heerwaarden, J., Assefa, T., Manners, R., Müller, A., Pè, M. E., Polar, V., Ramirez-Villegas, J., Solberg, S. Ø., Teeken, B., & Tufan, H. A. (2023). Data-driven approaches can harness crop diversity to address heterogeneous needs for breeding products. *Proceedings of the National Academy of Sciences of the United States of America*, *120*(14), 1–10. https://doi.org/10.1073/pnas.2205771120

FAO, ECA, & AUC. (2021). *Africa - Regional overview of food security and Nutrition: Statistics and Trends* (73 p.). FAO. https://doi.org/10.4060/cb7496en

Fischer, G., Gramzow, A., & Laizer, A. (2017). Gender, vegetable value chains, income distribution and access to resources: Insights from surveys in Tanzania. *European Journal of Horticultural Science*, *82*(6), 319–327. https://doi.org/10.17660/eJHS.2017/82.6.7

Fischer, G., Patt, N., Ochieng, J., & Mvungi, H. (2020). Participation in and gains from traditional vegetable value chains: A gendered analysis of perceptions of labour, income and expenditure in producers' and traders' households. *European Journal of Development Research*, *32*(4), 1080–1104. https://doi.org/10.1057/s41287-020-00257-0

Fredenberg, E., Karl, K., Passarelli, S., Porciello, J., Rattehalli, V., Auguston, A., Chimwaza, G., Grande, F., Holmes, B., Kozlowski, N., Laborde, D., MacCarthy, D., Masikati, P., McMullin, S., Mendez Leal, E., Valdivia, R., Van Deynze, A., & van Zonneveld, M. (2024). *Vision for Adapted Crops and Soils (VACS) Research in Action: Opportunity Crops for Africa*. https://doi.org/10.7916/3hd1-8t86

von Grebmer, K., Saltzman, A., Birol, E., Wiesmann, D., Prasai, N., Yin, S., Yohannes, Y., Menon, P., Thompson, J., & Sonntag, A. (2014). Global Hunger Index: The Challenge of Hidden Hunger. https://doi.org/10.4135/9781412995719.n46

Hammond, J., Rosenblum, N., Breseman, D., Gorman, L., Manners, R., van Wijk, M. T., Sibomana, M., Remans, R., Vanlauwe, B., & Schut, M. (2020). Towards actionable farm typologies: Scaling adoption of agricultural inputs in Rwanda. *Agricultural Systems*, *183*(November 2019), 102857. https://doi.org/10.1016/j.agsy.2020.102857

Hounsome, N., Hounsome, B., Tomos, D., & Edwards-Jones, G. (2008). Plant metabolites and nutritional quality of vegetables. *Journal of Food Science*, *73*(4), R48–R65. https://doi.org/10.1111/j.1750-3841.2008.00716.x

Jarvis, A., Gallo-Franco, J., Portilla, J., German, B., Debouck, D., Rajasekharan, M., … de la Parra, J. (2024). Periodic Table of food initiative for generating biomolecular knowledge of edible biodiversity. *Nature Food*, *5*(3), 189–193. https://doi.org/10.1038/s43016-024-00941-y

25722611, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1002/ppp3.70035 by Norwegian Institute Of Public Healt Invoice Receipt DFO, Wiley Online Library on [15/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Kachiguma, N. A., Mwase, W., Maliro, M., & Damaliphetsa, A. (2015). Chemical and mineral composition of Amaranth (Amaranthus L.) species collected from Central Malawi. *Journal of Food Research*, *4*(4), 92. https://doi.org/10.5539/jfr.v4n4p92

Kalmpourtzidou, A., Eilander, A., & Talsma, E. F. (2020). Global vegetable intake and supply compared to recommendations: A systematic review. *Nutrients*, *12*(6), 22–29. https://doi.org/10.3390/nu12061558

Kamga, R. T., Kouamé, C., Atangana, A. R., Chagomoka, T., & Ndango, R. (2013). Nutritional evaluation of five African indigenous vegetables. *Journal of Horticultural Research*, *21*(1), 99–106. https://doi.org/10.2478/johr-2013-0014

Kansiime, M. K., & Mastenbroek, A. (2016). Enhancing resilience of farmer seed system to climate-induced stresses: Insights from a case study in West Nile region, Uganda. *Journal of Rural Studies*, *47*, 220–230. https://doi.org/10.1016/j.jrurstud.2016.08.004

Kansiime, M. K., Ochieng, J., Kessy, R., Karanja, D., Romney, D., & Afari-Sefa, V. (2018). Changing knowledge and perceptions of African indigenous vegetables: The role of community-based nutritional outreach. *Development in Practice*, *28*(4), 480–493. https://doi.org/10.1080/09614524.2018.1449814

Keatinge, J. D. H., Wang, J. F., Dinssa, F. F., Ebert, A. W., Hughes, J. D. A., Stoilova, T., Nenguwo, N., Dhillon, N. P. S., Easdown, W. J., Mavlyanova, R., Tenkouano, A., Afari-Sefa, V., Yang, R. Y., Srinivasan, R., Holmer, R. J., Luther, G., Ho, F. I., Shahabuddin, A., Schreinemachers, P., … Ravishankar, M. (2015). Indigenous vegetables worldwide: Their importance and future development. *Acta Horticulturae*, *1102*(September), 1–20. https://doi.org/10.17660/ActaHortic.2015.1102.1

Kilwinger, F., Mugambi, S., Manners, R., Schut, M., Tumwegamire, S., Nduwumuremyi, A., Bambara, S., Paauwe, M., & Almekinders, C. (2021). Characterizing cassava farmer typologies and their seed sourcing practices to explore opportunities for economically sustainable seed business models in Rwanda. *Outlook on Agriculture*, *50*(4), 441–454. https://doi.org/10.1177/00307270211045408

Laajaj, R., Macours, K., Masso, C., Thuita, M., & Vanlauwe, B. (2020). Reconciling yield gains in agronomic trials with returns under African smallholder conditions. *Scientific Reports*, *10*(1), 14286. https://doi.org/10.1038/s41598-020-71155-y

Luce, R. D. (1959). *Individual choice behavior: A theoretical analysis*. Dover Publications. https://doi.org/10.1037/14396-000

Maechler, M., Rousseeuw, P., Struyf, A., Hubert, M., & Hornik, K. (2023). Cluster: Cluster analysis basics and extensions. R Package, v2.1.8.1. https://doi.org/10.32614/CRAN.package.cluster

McGuire, S., & Sperling, L. (2016). Seed systems smallholder farmers use. *Food Security*, *8*(1), 179–195. https://doi.org/10.1007/s12571-015-0528-8

McMullin, S., Stadlmayr, B., Mausch, K., Revoredo-Giha, C., Burnett, F., Guarino, L., Brouwer, I. D., Jamnadass, R., Graudal, L., Powell, W., & Dawson, I. K. (2021). Determining appropriate interventions to mainstream nutritious orphan crops into African food systems. *Global Food Security*, *28*, 100465. https://doi.org/10.1016/j.gfs.2020.100465

Mwadzingeni, L., Afari-Sefa, V., Shimelis, H., N'Danikou, S., Figlan, S., Depenbusch, L., Shayanowako, A. I. T., Chagomoka, T., Mushayi, M., Schreinemachers, P., & Derera, J. (2021). Unpacking the value of traditional African vegetables for food and nutrition security. *Food Security*, *13*(5), 1215–1226. https://doi.org/10.1007/s12571-021-01159-7

N'Danikou, S., de Sousa, K., van Zonneveld, M., Laporte, M.-A., Shango, A., Aglinglo, L. A., Legba, E. C., Houdegbe, A. C., Diarra, D., Dit, Y., Dolo, A., Sidibe, A., Ouerdraogo, O. C., Coulibaly, H., Enoch, G. A.-D., Kileo, A., Malulu, D., & Matumbo, Z. (2025). *Replication data for: Citizen science informs demand-driven breeding of opportunity crops (v1.0)*. Zenodo. https://doi.org/10.5281/zenodo.15204183

N'Danikou, S., van Zonneveld, M., Dinssa, F. F., Schafleitner, R., Harris, J., Schreinemachers, P., & Ramasamy, S. (2022). Mainstreaming African vegetables to improve diets and livelihoods. In S. Padulosi, E. D. I. O. King, D. Hunter, & M. S. Swaminathan (Eds.), *Orphan crops for sustainable food and nutrition security: Promoting neglected and underutilized species* (pp. 208–205). Routledge. https://cgspace.cgiar.org/bitstream/handle/10568/116467/book.pdf?sequence=3#page=235

Ochieng, J., Schreinemachers, P., Ogada, M., Dinssa, F. F., Barnos, W., & Mndiga, H. (2019). Adoption of improved amaranth varieties and good agricultural practices in East Africa. *Land Use Policy*, *83*(February), 187–194. https://doi.org/10.1016/j.landusepol.2019.02.002

Ojiewo, C., Keatinge, D. J. D. H., Hughes, J., Tenkouano, A., Nair, R., Varshney, R., Siambi, M., Monyo, E., Ganga-Rao, N., & Silim, S. (2015). The role of vegetables and legumes in assuring food, nutrition, and income security for vulnerable groups in sub-Saharan Africa. *World Medical and Health Policy*, *7*(3), 187–210. https://doi.org/10.1002/wmh3.148

Okello, J., Hutchinson, M., Mwang'ombe, A., Ambuko, J., Olubayo, F., & Mwakangalu, M. (2015). Consumer demand for value-added products of African indigenous vegetables in coastal Kenya: The case of sun-dried and frozen cowpea leaves. *Journal of Agriculture, Food Systems, and Community Development*, *6*(1), 189–207. https://doi.org/10.5304/jafscd.2015.061.004

Onim, M., & Mwaniki, P. (2008). Cataloguing and Evaluation of Available Community/Farmers-Based Seed Enterprises on African Indigenous Vegetables (AIVs) in Four EAC Countries. https://www.researchgate.net/profile/Kirigo-Phoebe/publication/346381263_Cataloguing_and_evaluation_of_available_communityfarmers-based_seed_enterprises_on_african_indigenous_vegetables_aivs_four_eca_countries/links/5fbf53d392851c933f5d28b0/cataloguing-and-evaluation-of-available-community-farmers-based-seed-enterprises-on-african-indigenous-vegetables-AIVs-four-ECA-countries.pdf, accessed 16 December 2024.

Plackett, R. L. (1975). The analysis of permutations. *Journal of the Royal Statistical Society Series C (Applied Statistics)*, *24*(2), 193–202. https://doi.org/10.2307/2346567

Quirós, C., de Sousa, K., Steinke, J., Madriz, B., Laporte, M.-A., Arnaud, E., Manners, R., Ortiz-Crespo, B., Müller, A., & van Etten, J. (2024). ClimMob: Software to support experimental citizen science in agriculture. *Computers and Electronics in Agriculture.*, *217*, 108539. https://doi.org/10.1016/j.compag.2023.108539

Schafleitner, R., Lin, Y. p., Dinssa, F. F., N'Danikou, S., Finkers, R., Minja, R., Abukutsa-Onyango, M., Nyonje, W. A., Lin, C. y., Wu, T. h., Sigalla, J. P., van Zonneveld, M., Hsiao, Y. y., Kumar, S., Wu, W. j., Wang, H. l., Lin, S., & Yang, R. y. (2022). The world vegetable center Amaranthus germplasm collection: Core collection development and evaluation of agronomic and nutritional traits. *Crop Science*, *62*(3), 1173–1187. https://doi.org/10.1002/csc2.20715

Schouten, H. J., Tikunov, Y., Verkerke, W., Finkers, R., Bovy, A., Bai, Y., & Visser, R. G. F. (2019). Breeding has increased the diversity of cultivated tomato in the Netherlands. *Frontiers in Plant Science*, *10*-(December), 1606. https://doi.org/10.3389/fpls.2019.01606

Schreinemachers, P., Howard, J., Turner, M., Groot, S. N., Dubey, B., Mwadzingeni, L., Chagomoka, T., Ngugi, M., Afari-Sefa, V., Hanson, P., & Wopereis, M. C. S. (2021). Africa's evolving vegetable seed sector: Status, policy options and lessons from Asia. *Food Security*, *13*(3), 511–523. https://doi.org/10.1007/s12571-021-01146-y

Schreinemachers, P., Simmons, E. B., & Wopereis, M. C. S. (2018). Tapping the economic and nutritional power of vegetables. *Global. Food Security*, *16*(September 2017), 36–45. https://doi.org/10.1016/j.gfs.2017.09.005

Stoilova, T., Van Zonneveld, M., Roothaert, R., & Schreinemachers, P. (2019). Connecting genebanks to farmers in East Africa through the distribution of vegetable seed kits. *Plant Genetic Resources: Characterisation and Utilisation*, *17*(3), 306–309. https://doi.org/10.1017/S1479262119000017

Tatsvarei, M. P., & Rukasha, S. (2022). Commercialization of indigenous vegetable value chains: A review of selected African countries "commercialization of indigenous vegetable value chains: A review of

25722611, 0, Downloaded from https://nph.onlinelibrary.wiley.com/doi/10.1002/ppp3.70035 by Norwegian Institute Of Public Healt Invoice Receipt DFO, Wiley Online Library on [15/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

selected African countries". *International Journal of Development and Sustainability*, *11*(6), 184–201. www.isdsnet.com/ijds

Thiele, G., Dufour, D., Vernier, P., Mwanga, R. O. M., Parker, M. L., Schulte Geldermann, E., Teeken, B., Wossen, T., Gotor, E., Kikulwe, E., Tufan, H., Sinelle, S., Kouakou, A. M., Friedmann, M., Polar, V., & Hershey, C. (2020). A review of varietal change in roots, tubers and bananas: Consumer preferences and other drivers of adoption and implications for breeding. *International Journal of Food Science and Technology*, *56*, 1076–1092. https://doi.org/10.1111/ijfs.14684

Turner, H. L., van Etten, J., Kosmidis, I., & Firth, D. (2020). Modelling rankings in R: the PlackettLuce package. In *Computational statistics* (Vol. 35, Issue 3) (pp. 1027–1057). Springer. https://doi.org/10.1007/s00180-020-00959-3

Turner, M. R., & Bishaw, Z. (2016). A Review of Variety Release Procedures and Related Issues with Recommendations for Good Practice (No. 31; Working Papers). https://hdl.handle.net/20.500.11766/6362

van Zonneveld, M., Kindt, R., McMullin, S., Achigan-Dako, E. G., N'Danikou, S., Hsieh, W., Lin, Y., & Dawson, I. K. (2023). Forgotten food crops in sub- Saharan Africa for healthy diets in a changing climate. *Proceedings of the National Academy of Sciences*, *120*(14), e2205794120. https://doi.org/10.1073/pnas

van Zonneveld, M., Kindt, R., Solberg, S., N'Danikou, S., & Dawson, I. K. (2021). Diversity and conservation of traditional African vegetables: Priorities for action. *Diversity and Distributions*, *27*(2), 216–232. https://doi.org/10.1111/ddi.13188

Voss, R. C., Cairns, J. E., Olsen, M., Muteti, F. N., Magambo Kanyenji, G., Hamadziripi, E., Ligeyo, D., Mashingaidze, K., Collinson, S., Wanderi, S., & Woyengo, V. (2023). Innovative approaches to integrating gender into conventional maize breeding: Lessons from the seed production Technology for Africa project. *Frontiers in Sociology*, *8*-(September), 1254595. https://doi.org/10.3389/fsoc.2023.1254595

Voss, R. C., Donovan, J., Rutsaert, P., & Cairns, J. E. (2021). Gender inclusivity through maize breeding in Africa: A review of the issues and options for future engagement. *Outlook on Agriculture*, *50*(4), 392–405. https://doi.org/10.1177/00307270211058208

Wanyama, R., Mvungi, H., Luoga, R., Mmasi, M., Zablon, E., N'Danikou, S., & Schreinemachers, P. (2023). Sustainability of one-time seed distributions: A long-term follow-up of vegetable seed kits in Tanzania. *Frontiers in Sustainable Food Systems*, *7*(2023), 1–10. https://doi.org/10.3389/fsufs.2023.1281692

Weinberger, K., & Pichop, G. N. (2009). Marketing of African Indigenous Vegetables along Urban and Peri-Urban Supply Chains in Sub-Saharan Africa. In C. M. Shackleton, M. W. Pasquini, & A. W. Drescher (Eds.), *African indigenous vegetables in urban agriculture* (pp. 225–244). Earthscan. https://cgspace.cgiar.org/server/api/core/bitstreams/31cbd46f-be77-43a2-b715-1daccbb972fa/content#page=266

Weltzien, E., Rattunde, F., Christinck, A., Isaacs, K., & Ashby, J. A. (2019). Gender and Farmer Preferences for Varietal Traits. In I. Goldman (Ed.), *Plant Breeding Reviews* (Vol. 43) (pp. 243–278). Wiley. https://doi.org/10.1002/9781119616801.ch7

Yang, R., & Keding, G. B. (2009). African Indigenous Vegetables in Urban Agriculture. In C. M. Shackleton, M. W. Pasquini, & A. W. Drescher (Eds.), *African indigenous vegetables in urban agriculture* (pp. 105–143). Routledge. https://doi.org/10.4324/9781849770019

Zeileis, A., Hothorn, T., & Hornik, K. (2008). Model-based recursive partitioning Torsten Hothorn. *Journal of Computational and Graphical Statistics*, *17*(2), 492–514. https://doi.org/10.1198/10618600SX319331

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